

Pandas : Introduction

- Pandas is a popular open-source data manipulation and analysis library for Python.
- It provides easy-to-use data structures like DataFrame and Series, which are designed to make working with structured data fast, easy, and expressive.
- Pandas are widely used in data science, machine learning, and data analysis for tasks such as data cleaning, transformation, and exploration.

What is the Pandas series?

- A Series is a one-dimensional, labeled array that can store a single column of any data type. You can think of it as a vector or a list with labels.
- A Series has an index, which can be labels, numbers, or a combination of both.
- You can create a Series from various sources, such as lists, dictionaries, scalars, or files.

What is Pandas Dataframe?

- A DataFrame is a two-dimensional, tabular data structure that can store multiple columns of different data types. You can think of it as a spreadsheet or a database table.
- A DataFrame has a row index and a column index, which can be labels, numbers, or a combination of both.
- You can create a DataFrame from various sources, such as lists, dictionaries, arrays, or files.

Pandas : Dataframe vs. Series

- One of the main differences between DataFrame and Series is that a DataFrame can have multiple columns, while a Series can only have one.
- This means that a DataFrame can store more complex and heterogeneous data, while a Series can store more simple and homogeneous data.
- Another difference is that a DataFrame can have different data types for each column, while a Series can only have one data type for the whole array.
- This means that a DataFrame can handle mixed data types, such as numbers, strings, booleans, or dates, while a Series can only handle one data type at a time.

Pandas : DataFrame vs. Series

Operations on DataFrame and Series

Series: -

- Mathematical or element-wise operations produce a new Series.
- The resulting Series incorporates the union of indexes from the original Series.
- Operations are executed based on matching labels.
- If there's no matching label between the two Series, the output value defaults to NaN.

DataFrame:

- Mathematical or element-wise operations generate a new DataFrame.
- The new DataFrame maintains the same columns and index as the originals.
- Computations occur element-wise across corresponding positions.
- It facilitates operations on multiple variables concurrently, such as filtering rows and selecting specific columns.

Pandas : Dataframe vs. Series

Series	DataFrame
One- dimensional	Two- dimensional
Series elements must be homogenous.	Can be heterogeneous.
Immutable(size cannot be changed).	Mutable(size can be changeable).
Element wise computations.	Column wise computations.
Functionality is less.	Functionality is more.
Alignment not supported.	Alignment is supported.

Summary

- Pandas offers two vital data structures, Series and DataFrame, each tailored for specific data manipulation tasks.
- Series excel in handling one-dimensional labeled data with efficient indexing and vectorized operations.
- DataFrames provide tabular data organization with versatile indexing, column operations, and robust handling of missing data.